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# Bibliography

1. “Design Patterns.” *Refactoring.Guru*, https://refactoring.guru/design-patterns. Accessed 20 October 2025.
    - Used it to understand the concept of Interfaces and its usage. Resulted in the current class structure of lifelines.
    - This was recommended to us by our tutor.
2. MillionaireSound. “MillionaireSound.” *YouTube Channel of MillionaireSound*, https://www.youtube.com/@MillionaireSound. Accessed 24 10 2025.
    - This is the YouTube channel that provided the sound effects for the game.
    - It helped with user experience, allowing us to bring a feel of the real life TV show to the game.
    - This was found by searching YouTube for inspiration for the game.
3. Oracle. “The Java Tutorials.” *ORACLE*, https://docs.oracle.com/javase/tutorial/uiswing/events/eventsandcomponents.html. Accessed 20 October 2025.
    - To know about the Jswing and atw components and its specific commands. Like setting borders for the Jbutton, setting custom font, etc.
    - Our manual for most of the GUI resulting in the user Interface we have currently.
    - This was shared with us in lectures.
4. RyiShow. “Create Timer - Countdown/Normal/Two Digit - Java Extra 32.” *YouTube*, 2020, https://www.youtube.com/watch?v=zWw72j-EbqI. Accessed 18 October 2025.
    - Tutorial on how to add a timer to code. This helped create the timer on the question screen.

- It was the best video tutorial that met the needs of the timer for the project when "how to create a timer in Java" on YouTube.

5. Roach, Claire. "Who Wants To Be A Millionaire Quiz Questions & Answers 2025." *Quiz Questions UK*, https://quiz-questions.uk/who-wants-to-be-a-millionaire/. Accessed 15 October 2025.
    - We searched for questions that were usually used in the live TV show that we could use for this game as well and came across this website that had the question bank from the 2025 season of the show.
6. Sinha, Shiksha. "How to add audio on JSwing in Java." *CodeSpeedy*, https://www.codespeedy.com/how-to-add-audio-on-jswing-in-java/. Accessed 24 10 2025.
    - A website that provides a guide on how to add audio on JSwing in Java.
    - Found through research.
7. Geeksforgeeks Tutorial. "Java Tutorial" GEEKSFORGEEKS, https://www.geeksforgeeks.org/java/java/. Accessed 15 10 2025.
    - A highly useful resource that was recommended in high school, commonly used for coding tutorials.
    - The website provided us with tutorials on how to utilize OOP structures, reading CSV files, and other elements of Java that we needed to build the game.

# Learnings

While developing Who Deserves to Be a Millionaire, we gained valuable experience developing and enhancing graphical user interfaces (GUIs) with Java Swing. By focusing on how the layout, colors, type, and organization of components can impact usability and interaction with players, we played with a variety of GUI functions, including layout managers, border factory, colours etc to organize windows, buttons, and panels. This ultimately allowed us to create an organized, clear, and readable interface to communicate information, including questions, answers, lifelines, and timers, without overwhelming the user. We also picked up how to add event listeners to create an interface that is responsive and dynamic—so that buttons, timers, and pop-up windows all respond immediately to user actions. When we created screens that retained a uniform visual theme and added graphics, such as the road map panel and pop-ups at the end of games, we learned how important it is to maintain design consistency and visual flow. But we didn't just learn aesthetic considerations; we also learned that visual feedback, such as disabling lifelines that had been used, highlighting levels in the roadmap, and appropriate sound effects, helped to enhance the player's sense of agency and immersion into the game world.

In addition to GUI development, we gained much experience working with external resources and built-in functionalities in Java. We learned about importing and using sound, fonts, images, and text files into our project, adding another level of depth and realism to the gameplay experience. Working with sound helped us learn how to playback audio while not interrupting the user interface. Using custom fonts and themed images worked to create the TV-show-like atmosphere we desired. We also learned how to effectively manage file imports. For example, we created and loaded sets of questions based on a given level which allowed us to improve scalability and organization. Another learning point was randomization- both in our choice of

questions and the addition of random events occurring in-game. Using Java's Random class in this way meant that we could add elements of unpredictability through event cards that altered gameplay both positively and negatively. This also strengthened our understanding of how controlled randomness could make a game more dynamic and replayable. Overall, this project has strengthened our technical and design skills to visualize how to incorporate multiple programming concepts, into the same cohesive and engaging application.